Assignment – Data Science (End-to-End Time Series Forecasting System with API) Objective:

Build a **production-ready forecasting system** that:

1. Trains **multiple forecasting algorithms**
2. Compares and selects the **best model**
3. Exposes predictions via a **REST API**
4. Should be designed like real backend service

Dataset:

Use the attached data set for your case study

Shared in [excel file]

Problem Statement

Forecast **next 8weeks of sales** for each state using historical data. Your solution must:

- Handle missing dates / missing values (if any)
- Handle seasonality & trend
- Automatically select the best performing model
- Serve predictions via API

Mandatory Models to Implement

Train and compare at least:

1. **ARIMA** / **SARIMA**
2. **Facebook Prophet**
3. **XGBoost** (with lag features)
4. **LSTM** (deep learning)

Feature Engineering (critical part)

You must create:

- Lag features (t-1, t-7, t-30)
- Rolling mean / std
- Day of week, month, holiday flag
- Train / validation split using time series logic (no leakage)

Create a short video of your solution and share it along with Code and documentation